



POLITECNICO
MILANO 1863

Dynamics of Electrical Machines and Drives

Clarke and Park transformations

Matteo Benvenuti Matteo Sposito Andrea Polastri Mattia Rossi

Three phase system

Quantities

- We start by introducing **three-phase systems** with alternating currents
- Consider a *balanced three-phase voltage source*, the electrical variable in each phase are sinusoidal with the peak value $\sqrt{2}X_{\text{rms}}$ where X_{rms} is the rms value and $\omega = 2\pi f$ is the angular frequency

$$x_a(t) = \sqrt{2}X_{\text{rms}} \sin(\omega t)$$

$$x_b(t) = \sqrt{2}X_{\text{rms}} \sin\left(\omega t - \frac{2}{3}\pi\right)$$

$$x_c(t) = \sqrt{2}X_{\text{rms}} \sin\left(\omega t + \frac{2}{3}\pi\right)$$

$$x_{ab}(t) = x_a(t) - x_b(t)$$

$$x_{bc}(t) = x_b(t) - x_c(t)$$

$$x_{ca}(t) = x_c(t) - x_a(t)$$

- Their phase are shifted by $2\pi/3$ each other
- **Variables:** $v(t)$, $i(t)$

Stationary Reference Frame

$\alpha\beta 0$ model

- All variables from three-phase balanced abc system are transformed to a stationary orthogonal reference frame $\alpha\beta 0$

$$\mathbf{K} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad \text{or} \quad \mathbf{K}^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 1 \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 1 \end{bmatrix}$$

- The factor $\frac{2}{3}$ ensures that the amplitudes of the three-phase variables are preserved, **amplitude invariant**
- Changing the factor from $\frac{2}{3}$ to $\sqrt{\frac{2}{3}}$ ensures **power invariant**

Rotating Reference Frame

dq model

- φ is the angle between d -axis and a -axis $\varphi(t) = \int_0^t \omega_{\text{ref}}(\tau) d\tau + \varphi(0)$ and ω_{ref} defines the angular speed of the reference frame
- The so-called **Park transformation** maps the vector $\xi_{abc} = [\xi_a \ \xi_b \ \xi_c]^T$ to the vector $\xi_{dq0} = [\xi_d \ \xi_q \ \xi_0]^T$, and vice versa

$$\xi_{dq0} = \mathbf{K}(\varphi) \xi_{abc} \quad \text{or} \quad \xi_{abc} = \mathbf{K}^{-1}(\varphi) \xi_{dq0}$$

where

$$\mathbf{K}(\varphi) = \frac{2}{3} \begin{bmatrix} \cos \varphi & \cos(\varphi - \frac{2}{3}\pi) & \cos(\varphi + \frac{2}{3}\pi) \\ -\sin \varphi & -\sin(\varphi - \frac{2}{3}\pi) & -\sin(\varphi + \frac{2}{3}\pi) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\mathbf{K}^{-1}(\varphi) = \begin{bmatrix} \cos \varphi & -\sin \varphi & 1 \\ \cos(\varphi - \frac{2}{3}\pi) & -\sin(\varphi - \frac{2}{3}\pi) & 1 \\ \cos(\varphi + \frac{2}{3}\pi) & -\sin(\varphi + \frac{2}{3}\pi) & 1 \end{bmatrix}$$

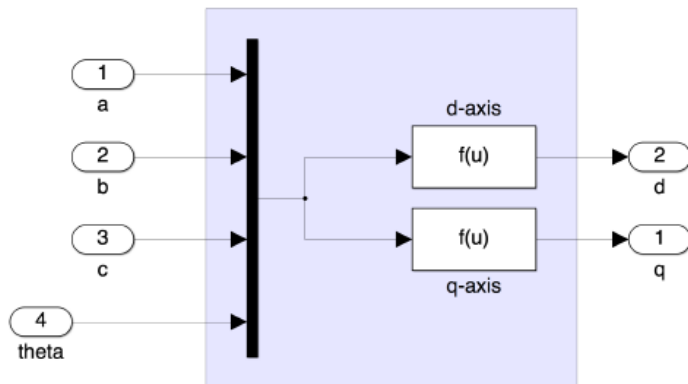
- **Synchronous reference frame** where the d -axis is aligned with stator or rotor flux vector and by setting ω_{ref} to the flux's angular speed ω_s

Rotating Reference Frame

Simulink

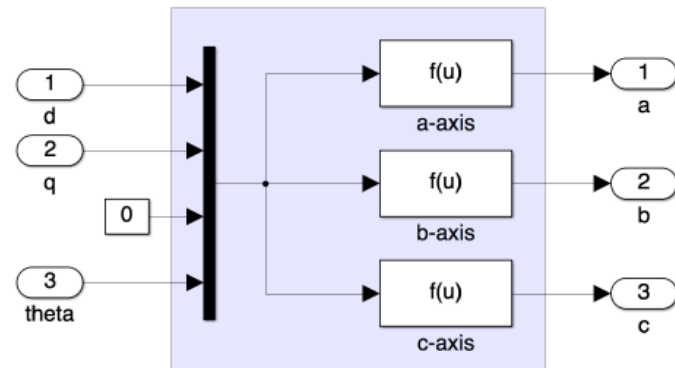
Park transformation:

$$\xi_{dq0} = K(\varphi)\xi_{abc}$$



Park anti-trasformation:

$$\xi_{abc} = K^{-1}(\varphi)\xi_{dq0}$$



- d-axis: $f(u) = \sqrt{\frac{2}{3}}(u[1]\cos(u[4]) + u[2]\cos(u[4] - \frac{2\pi}{3}) + u[3]\cos(u[4] + \frac{2\pi}{3}))$
- q-axis: $f(u) = \sqrt{\frac{2}{3}}(-u[1]\sin(u[4]) - u[2]\sin(u[4] - \frac{2\pi}{3}) - u[3]\sin(u[4] + \frac{2\pi}{3}))$

Exercise

Clarke/Park subsystems

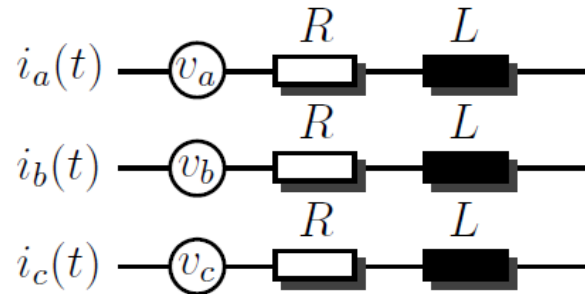
Built a Simulink subsystem for the previous Park transformations:

- create two subsystems for Clarke/Park Transformations and test ac quantities: 100V (rms) at 50Hz
- compare your Clarke/Park blocks with Built-in Simulink blocks

RL model

Now with a sinusoidal supply

Three-phase RL load:



$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$

Remember

- voltage and current components: $\mathbf{v}_{dq0} = \mathbf{K}(\varphi) \mathbf{v}_{abc}$
- resistance and inductance: $\mathbf{L}_{dq0} = \mathbf{K}(\varphi) \mathbf{L}_{abc} \mathbf{K}^{-1}(\varphi)$

- **Clarke transformation:**

$$\begin{bmatrix} v_\alpha \\ v_\beta \\ v_0 \end{bmatrix} = \begin{bmatrix} L-M & 0 & 0 \\ 0 & L-M & 0 \\ 0 & 0 & L+2M \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_\alpha \\ i_\beta \\ i_0 \end{bmatrix} + \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \\ i_0 \end{bmatrix}$$

- **Park transformation:**

$$\begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix} = \begin{bmatrix} L-M & 0 & 0 \\ 0 & L-M & 0 \\ 0 & 0 & L+2M \end{bmatrix} \left(\frac{d}{dt} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} + \underbrace{\omega J \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix}}_{\text{motional term}} \right) + \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix}$$

$$\text{where } J = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } \omega = \frac{d\varphi}{dt}$$

Exercise

Three-phase RL model

Built a Simulink scheme of the previous RL model:

- create a subsystem in dq component
 - input: voltages, angle φ
 - output: currents
- data: $L = 100mH$, $M = 80mH$, $R = 2\Omega$, $V_{rms} = 100V$, $f = 50Hz$
- compare the Clarke/Park effects on the phase voltages/currents
- plot the $\alpha\beta$ and dq plot for voltages/currents

Hint:

- start from dq -model and then play with to obtain $\alpha\beta$ model
- in Simulink, start to built a path from the desired output: (example)

$$\frac{di_d}{dt} = \frac{1}{L}v_d - \frac{R}{L}i_d + \omega i_q \rightarrow i_d = \underbrace{\int}_{1/s} \left(\frac{1}{L}v_d - \frac{R}{L}i_d + \omega i_q \right) dt$$